



## FINAL EXAMINATION SEMESTER III, SESSION 2020/2021

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COURSE CODE : FSPP 0034  
COURSE : PHYSICS 1I  
PROGRAMME : FOUNDATION UTM  
DURATION : 3 HOURS  
DATE : 11 APRIL 2021

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### INSTRUCTION TO CANDIDATES:

1. ANSWER ALL QUESTIONS.

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### **WARNING!**

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This examination paper consists of (12) printed pages only including this page

# PART B

**QUESTION 4 (20 MARKS)**

- a) In Young's double slit experiment, the slit separation is 0.25 mm and the screen is 0.15 m from the slits. When the slits are illuminated by a light, the third dark fringes are observed 9.5 mm from the central bright fringe. Calculate:
- The wavelength of the light.  $6.33 \times 10^{-6} \text{ m}$
  - The separation between two consecutive fringes.  $3.8 \times 10^{-3} \text{ m}$
  - How far the screen should be moved from its original position if the fringe separation is changed to 5.50 mm and the other parameters are maintained.  $0.07 \text{ m}$

(6 marks)

- b) Light from sodium lamp of wavelength 580 nm is shined directly on a narrow slit with a width of 0.02 mm. Determine the angular width of the central bright fringe.  $3.32$

$$\theta_w = 2\theta_1 = 2\left(\sin^{-1}\frac{\lambda}{a}\right) \quad w = 2y_1 = \frac{2\lambda L}{a} \quad (2 \text{ marks})$$

- c) The fourth order line for 650 nm light falling on the diffraction grating is observed at  $15.5^\circ$  angle. Determine:

- The number of lines per cm for the grating.  $1027.8 \text{ line/cm}$   $d = \frac{1}{N}$
- The angle of the second order line.  $7.7$

(5 marks)

- d) When an electromagnetic radiation of wavelength 500 nm is incident on a metal surface, photoelectrons are emitted. If the photoelectrons are stopped by a voltage of 2.0 V, calculate:

- The work function of this metal in eV.  $0.47 \text{ eV}$
- The de Broglie wavelength of the electrons.  $5.68 \times 10^{-10} \text{ m}$

(7 marks)

4) (a) (i)  $x_m = (m + \frac{1}{2}) \frac{\lambda}{d}$  (ii)  $\Delta y = \frac{\lambda L}{d} = \frac{6.33 \times 10^{-6} (0.15)}{0.25 \times 10^{-3}}$  (iii)  $5.5 \times 10^{-3} = \frac{6.33 \times 10^{-6} (L)}{0.25 \times 10^{-3}}$

$9.5 \times 10^{-3} = (2 + \frac{1}{2}) \frac{\lambda (0.15)}{0.25 \times 10^{-3}}$

$9.5 \times 10^{-3} = 1500 \lambda$

$\lambda = 6.33 \times 10^{-6} \text{ m}$

$= 3.8 \times 10^{-3} \text{ m}$

$= 0.217$

$0.217 - 0.15 = 0.067 \text{ m}$

(b)  $\lambda = 580 \times 10^{-9} \text{ m}$ ,  $a = 0.02 \times 10^{-3} \text{ m}$

$\theta = 2 \sin^{-1} \left( \frac{\lambda}{a} \right)$

$= 2 \sin^{-1} \left( \frac{580 \times 10^{-9}}{0.02 \times 10^{-3}} \right)$

$= 3.32^\circ$

(c) (i)  $KE = PE$

$E = W_0 + K_{\text{max}}$

$W_0 = E - K_{\text{max}}$

$W_0 = E + PE$

$= \frac{hc}{\lambda} + eV$

$= \frac{6.63 \times 10^{-34} (3 \times 10^8)}{500 \times 10^{-9}} + 1.6 \times 10^{-19} (2) = 0.49 \text{ eV}$

(ii)  $\frac{1}{2} mv^2 = eV$   $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} (8.38 \times 10^5)}$

$v = 8.38 \times 10^5 \text{ m/s}$   $= 8.68 \times 10^{-10} \text{ m}$

(d)  $\lambda = 650 \text{ nm}$ ,  $\theta = 15.5$ ,  $n = 4$

(i)  $d \sin \theta = n \lambda$

$d = \frac{1}{N}$

$d = \frac{4 (650 \times 10^{-9})}{\sin (15.5)}$

$N = \frac{1}{9.73 \times 10^{-6}}$

$= 9.73 \times 10^{-6} \text{ m}$

$= 102784 \text{ lines per meter}$

(ii)  $d \sin \theta = n \lambda$

$9.73 \times 10^{-6} \sin \theta = 2 (650 \times 10^{-9})$

$\sin \theta = 0.1336$

$\theta = 7.68^\circ$

**QUESTION 5 (20 MARKS)**

- a) Radioactive decay is a random and spontaneous nuclear reaction. Explain the terms random and spontaneous.

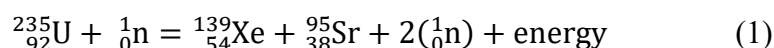
(2 marks)

- b) A laboratory has 2.56  $\mu\text{g}$  of pure  $^{214}_{83}\text{Bi}$ , which has a half-life of 20 min. Determine:

- The number of nuclei present initially.  $7.2 \times 10^{15}$
- The initial activity.  $4.16 \times 10^{12}$
- The activity after 1.00 hr.  $5.19 \times 10^{11}$
- The time,  $t$  when  $\left|\frac{dN}{dt}\right| = 1.00 \text{ s}^{-1}$ .  $13.96\text{h}$

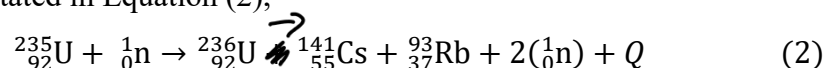
(8 marks)

- c) Equation (1) describe the nuclear fission reaction of Uranium-235 nuclei by slow moving neutrons. Calculate the energy released in the reaction.  $211 \text{ MeV}$



(2 marks)

- d) An atomic bomb uses 180 kg of U-235. If all of U-235 in the atomic bomb undergo fission as stated in Equation (2),

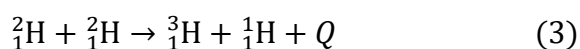


Determine:

- The mass loss in the fission reaction.  $0.19405 \text{ u}$
- The total reduction of mass of U-35 in atomic bomb.  $8.95 \times 10^{25} \text{ u}$

(4 marks)

- e) i. Why does a fusion reaction take place at high temperature?
- ii. Two deuteron fuses together to form a triton and a proton as stated in Equation (3). Calculate the energy released.  $4.03 \text{ MeV}$



(4 marks)

$$5) (b) (i) \quad 214g \Rightarrow 6.02 \times 10^{23} \text{ nuclei}$$

$$1.56 \times 10^{-6}g \Rightarrow 7.201 \times 10^{15} \text{ nuclei (N)}$$

$$(ii) \quad T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda} \quad A_0 = \lambda N_0$$

$$\lambda = 5.78 \times 10^{-4} \text{ s}^{-1} \quad = 5.78 \times 10^{-4} \times 7.201 \times 10^{15}$$

$$= 4.16 \times 10^{12} \text{ d/s}$$

$$(iii) \quad A = A_0 e^{-\lambda t}$$

$$= 4.16 \times 10^{12} \times e^{-(5.78 \times 10^{-4})(60 \times 60)}$$

$$= 5.193 \times 10^{11} \text{ d/s}$$

$$(iv) \quad A = \left| \frac{dN}{dt} \right| = 1.0 \quad 1730 = 7.2 \times 10^{15} e^{-5.78 \times 10^{-4} \times t}$$

$$5.78 \times 10^{-4} = \frac{\frac{dN}{dt}}{N} \quad \ln(2402 \times 10^{-16}) = -5.78 \times 10^{-4} t$$

$$N = 1730 \quad -24.06 = -5.78 \times 10^{-4} t$$

$$t = 50271 \text{ s} = 13.964 \text{ hour}$$

$$(c) \quad Q = (235 \times 7.60) - (139 \times 8.39 + 95 \times 8.74)$$

$$= 1786 - 1996.51$$

$$= -210.51 \text{ MeV}$$

$$(d) (i) \quad \Delta m = (235.04392 + 1.00867) - (140.91963 + 92.92157 + 2(1.00867))$$

$$= 0.194054 \text{ (mass defect for one nucleus U-235)}$$

$$(ii) \quad 0.235 \text{ kg} \rightarrow 6.023 \times 10^{23} \text{ nuclei}$$

$$180 \text{ kg} \rightarrow 4.613 \times 10^{26} \text{ nuclei}$$

$$\therefore \text{mass defect for } 180 \text{ kg U-235} = 0.194054 \times 4.613 \times 10^{26} = 8.95 \times 10^{25} \text{ u}$$

(e) (i) combine two small nuclei to form a larger nucleus

$$(ii) \quad \Delta m = m_i - m_p$$

$$= (2.014102 \times 2.014102) - (3.016049 + 1.007825)$$

$$= 4.33 \times 10^{-3} \text{ u}$$

$$Q = \Delta m c^2$$

$$= (4.33 \times 10^{-3} \times 931.5) (3 \times 10^8)^2$$

$$= 4.03 \text{ MeV}$$

## APPENDIX 1

**List of constant**

$$p = +1.60 \times 10^{-19} \text{ C}$$

$$e = -1.60 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$g = 9.80 \text{ m s}^{-2}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$k = 9.0 \times 10^9 \text{ N m}^2 \text{C}^{-2}$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$N_A = 6.022 \times 10^{23}$$

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = 935 \text{ MeV}/c^2$$

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

$$\text{Resistivity of iron at } 20^\circ \text{C: } 9.71 \times 10^{-8} \Omega \text{m}$$

$$\text{Resistivity of tungsten at } 20^\circ \text{C: } 5.6 \times 10^{-8} \Omega \text{m}$$

$$\text{Resistivity of platinum at } 20^\circ \text{C: } 10.6 \times 10^{-8} \Omega \text{m}$$

$$\text{Temperature coefficient of iron, } \alpha: 0.0651^\circ \text{C}^{-1}$$

$$\text{Temperature coefficient of tungsten, } \alpha: 0.045^\circ \text{C}^{-1}$$

$$\text{Temperature coefficient of platinum, } \alpha: 0.00397^\circ \text{C}^{-1}$$

## APPENDIX 2

## List of Formula

$$E = k \frac{Q}{r^2}$$

$$F = k \frac{Q_1 Q_2}{r^2}$$

$$U = k \frac{qQ}{r}$$

$$V = k \frac{Q}{r}$$

$$W = qV$$

$$\Delta U = q \Delta V$$

$$\Delta V = -Ed$$

$$F = qvB \sin \theta$$

$$B = \frac{\mu_0 I}{2\pi d}$$

$$F = ILB \sin \theta$$

$$U = \frac{1}{2} LI^2$$

$$P = IV$$

$$\rho_T = \rho_o [1 + \alpha(T - T_o)]$$

$$j = \frac{I}{A}$$

$$I = \frac{Q}{t}$$

$$M = \frac{\mu_o N_1 N_2 A}{l}$$

$$L = \frac{\mu_o N^2 A}{l}$$

$$I = I_{max} (1 - e^{-t/\tau}) = \frac{V}{R} (1 - e^{-t/\tau})$$

$$Q = Q_o e^{-t/\tau}$$

$$Q = Q_o (1 - e^{-t/\tau})$$

$$I = I_o e^{-t/\tau}$$

$$\varepsilon = -L \frac{dI}{dt}$$

$$\tau = \frac{L}{R}$$

$$\varepsilon = -\frac{d\phi_B}{dt}$$

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

$$v = f\lambda$$

$$E = Bv$$

$$k = \frac{2\pi}{\lambda}$$

$$P = \frac{1}{f}$$

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{f} = (n-1) \left[ \frac{1}{r_1} + \frac{1}{r_2} \right]$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$d \sin \theta = m\lambda, m = 0, 1, 2, 3 \dots$$

$$d \sin \theta = \left( m + \frac{1}{2} \right) \lambda, m = 0, 1, 2, 3 \dots$$

$$Q = \Delta m c^2$$

$$KE = \frac{hc}{\lambda} - W_o$$

$$\left| \frac{dN}{dt} \right|_o = \lambda N_o$$

$$\left| \frac{dN}{dt} \right| = \left| \frac{dN}{dt} \right|_o e^{-\lambda t}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\varepsilon = \left| -M \frac{dI}{dt} \right|$$

$$I = \frac{\varepsilon}{R}$$



$$\varepsilon = NB\omega A$$

$$\varepsilon = Blv$$

$$B_o = \frac{E_o}{c}$$

### APPENDIX 3

| Nuclide                | Binding energy per nucleon, $E$ (MeV) |
|------------------------|---------------------------------------|
| $^{95}_{38}\text{Sr}$  | 8.74                                  |
| $^{139}_{54}\text{Xe}$ | 8.39                                  |
| $^{235}_{92}\text{U}$  | 7.60                                  |

| Nuclide                | Binding energy per nucleon, $E$ (J) |
|------------------------|-------------------------------------|
| $^{144}_{56}\text{Ba}$ | $1.3341 \times 10^{-12}$            |
| $^{90}_{36}\text{Kr}$  | $1.3864 \times 10^{-12}$            |
| $^{235}_{92}\text{U}$  | $1.2191 \times 10^{-12}$            |

| Nuclide                | Mass, $m$ (u) |
|------------------------|---------------|
| $^{235}_{92}\text{U}$  | 235.04392     |
| $^{141}_{55}\text{Cs}$ | 140.91963     |
| $^{93}_{37}\text{Rb}$  | 92.92157      |
| $^1_0\text{n}$         | 1.00867       |
| $^1_1\text{H}$         | 1.007825      |
| $^2_1\text{H}$         | 2.014102      |
| $^3_1\text{H}$         | 3.016049      |
| $^3_2\text{He}$        | 3.01605       |

